



Infrastructure Resilience to Climate Change

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[https://doi.org/10.1061/\(ASCE\)IS.1943-555X.0000532](https://doi.org/10.1061/(ASCE)IS.1943-555X.0000532)

The special collection on Infrastructure Resilience to Climate Change can be found in the ASCE Library (https://ascelibrary.org/page/jitse4/infrastructure_resilience_climate_change).

There is a scene in the 1954 movie *Sabrina*, starring Humphrey Bogart and Audrey Hepburn, where Humphrey Bogart's character, Linus Larrabee, has many people jumping on a slab of clear plastic to jokingly show David Larrabee (William Holden) how resilient the slab is. The definition of the term *resilience* has most certainly dramatically changed since 1954. From materials science, it has pervaded in one form or another nearly all disciplines in the sciences, social sciences, and engineering. Holling (1973) is often credited as a pioneer of applying *resilience* away from its initial materials context to the study of the dynamics of ecosystems.

The term has taken on greater significance with growing evidence of the potentially monumental hazards from climate change. In the engineering realm, many recognize Hurricane Katrina, which killed nearly 2,000 people in the New Orleans region in August 2005, as a turning point. Until then, many assumed that threats from climate change would manifest themselves slowly and steadily (e.g., sea-level rise) along with slowly rising temperatures. But there is now strong evidence that suggests that climate change has a real and significant impact on the frequency and magnitude of extreme weather events (US Global Change Research Program 2009; National Climate Assessment 2018), such as Hurricane Katrina, that seem evermore present; from hurricanes and typhoons to tornadoes and ice storms, weather-related disasters are common in the news media. In engineering, increasing attention is being paid to designing systems with characteristics that range from sensing, anticipating, adapting, and learning (SAAL) (Park et al. 2013) to minimizing the consequences of failure (i.e., safe to fail) (Ahern 2011). Resilience is also about multifunctionality and balancing social/institutional, ecological, and technological dimensions, to name a few.

Redefining resilience is therefore paramount, with special relevance to the civil engineering field. The main objective of this special collection is to offer substantive studies that analyze and discuss infrastructure resilience within the global context of a changing climate. Purposefully, the special collection is designed to break free of the traditional disciplinary boundaries, involving all civil infrastructure systems, with works that can be theoretical or/and empirical. After all, whether it is water/wastewater systems,

transportation networks [including bridges (Saeidpour et al. 2018) and airports (Ryerson 2018)], power and natural gas systems (Portante et al. 2017), or building stock (Nahlik et al. 2017), all are currently vulnerable to some degree to climate change. Moreover, although most articles are expected to be heavily technical by nature—for example, Bristow and Hay (2017), Pozzi et al. (2017), and Xie et al. (2017)—we are also pleased to offer more qualitative contributions that discuss how formal protocols and institutional change can drive society to become more resilient, such as Zimmerman et al. (2017), Douglas et al. (2017), and Uda and Kennedy (2018). Finally, the most frequent hazard studied, with five contributions out of fourteen, is flooding, as it has plagued cities and regions across the United States and the world, whether from precipitation (Cook et al. 2017; Pregolato et al. 2017; Wisetjindawat et al. 2017) or sea-level rise (Sadler et al. 2017).

Overall, the realm of infrastructure resilience, especially in civil engineering, is still in its infancy. This special collection in the *ASCE Journal of Infrastructure Systems* should serve as an initial platform for the exchange of lessons learned and discoveries made. We are excited to see how the discussion will carry on.

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